

A Small Touch-Controlled SOLID STATE RELAY

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The demand for modern touch-based control systems is rising quickly, driven by their sleek design, ease of use, and lack of mechanical components. This system introduces an AC touch-controlled solid-state relay (SSR) capable of handling input voltages from 100V to 250V AC. It is designed to switch loads such as bulbs, fans, and other household appliances. At its core is the TTP223 touch sensor IC, which integrates an ADC converter and supports selectable touch modes via a solder jumper (J1). These modes can be configured through TOG (toggle) and AHLB (active high/low at power-up) settings, offering flexibility in operation.

The switching is managed by a BT136S TRIAC, delivering reliable

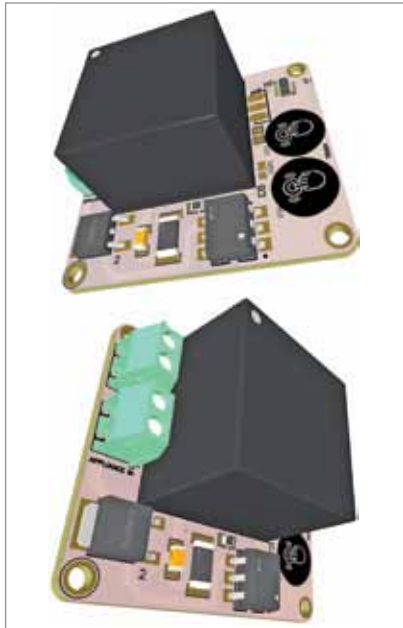


Fig. 1: Touch solid-state relay (SSR) PCB

SSR performance without moving parts. By combining touch sensing and relay functions, the system achieves a compact, smart, and silent design within a 4cm footprint. Two terminal inputs provide straightforward connections—one to the AC power line and the other to the appliance, ensuring simple and efficient control.

Fig. 1 shows a compact, modern AC solid-state relay (SSR) module with integrated touch controls. It features two capacitive touchpads (clearly marked with touch icons), driven by the TTP223 IC, allowing connected appliances to be toggled with a simple tap. The large black component is an optically isolated TRIAC-based SSR or relay, designed for switching high-voltage AC loads safely.

The BT136S TRIAC serves as the

BILL OF MATERIALS

Name	Desinator	Footprint	Quantity	Manuf. Part	Manuf.	Suppl.part
C-0805	C1	CAPACITOR_0805	1			
30pF	C2	201	1	GJM0335C0J300JB01D	Murata Electronics	81-GJM-0335C0J300JB1D
100nF	C3	C1206	1	CC1206KKX7RYBB104	YAGEO	C106241
Toggle Mode Jumper	J1	JUMPER_SOLDERPAD_2-1	1			
LED	LED1	LED_CHIP-LED0805	1			
1kΩ	R1	R0805	1	0805W8F1002T5E	UNI-ROYAL	C17414
220Ω	R2	R0402	1	RC0402FR-07220RL	YAGEO	C112291
100Ω	R3	RES-SMD_L6.4-W3.2-R2512	1	SR2512FK-7W100RL	YAGEO	C876841
BT136S	U1	T0-252-2_L6.6-W6.1-P4.57-LS9.9-BR-CW	1	BT136S-600D	GOODWORK	C18186224
MOC3041SM_C22373366	U2	SMD-6_L8.8-W6.4-P2.54-LS9.4-BL	1	MOC3041SM	AOTE	C22373366
LD05-23B05R2_C42432360	U3	PWRM-TH_LD05-23BXXR2	1	LD05-23B05R2	LZTEC	C42432360
TTP223 TOUCH PAD	U4	TOUCH_PAD	2			
TTP223E-BA6	U5	SOT-23-6_L2.9-W1.6-P0.95-LS2.8-BR	1	TTP223E-BA6	TONTEK	C129457
3pin_solder_pad	U6	3PIN_SOLDER_PAD	1			
DB301V-3.5-2P-GN	U7, U8	CONN-TH_DB301V-3.5-2P-GN	2	DB301V-3.5-2P-GN-S	DORABO	C695629